# 3 BJT DC Biasing

## BIPOLAR JUNCTION TRANSISTOR DC BIASING

- In the course of electronic devices, we have studied the construction and working of BJT and how P-type and N-type semiconductor materials are fabricated together to form a transistor.
- In this chapter, we will discuss how BJT behaves when we apply some external input signal and which configuration of BJT behaves more efficiently.
- In this chapter, we will deal with the addition of resistor and other passive elements to the transistor so that it can be used as an amplifier or a switch.
- Transistors with proper biasing can be used in many applications.
- The two types of signals applied to the BJT are-
  - AC signal
  - DC signal
- DC signal is mainly for the biasing of the transistor and acts as a power supply.
- The AC signal is the primary signal to be amplified using the transistor. It is usually the message signal.
- Let us revise what we learned in electronic devices and circuits about BJT.
- Transistor is a three-layer device formed by two semiconductors of the same type and one semiconductor of a different type, i.e. either two N-type and one P-type or two P-type and one N-type semiconductor material.
- One type of semiconductor is sandwiched between other similar types of semiconductors.
- When an N-type semiconductor is sandwiched between P-type semiconductors, then we get a PNP transistor, as shown below.



**Rack Your Brain**

Why is BJT called a bipolar device?

**Grey Matter Alert!**

Transistor is the fundamental building block of computers and other modern electronic devices.

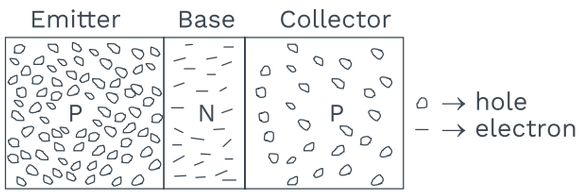



**Fig. 3.1 PNP Transistor**

- When a P-type semiconductor is sandwiched between two N-type semiconductors, then we get NPN transistor, as shown below.

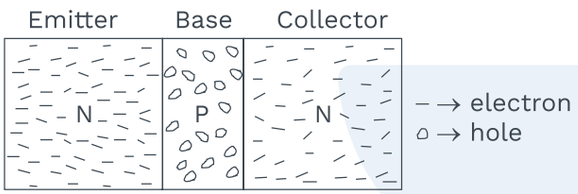



**Fig. 3.2 NPN Transistor**

- As shown in the above transistor structure, there are three regions called emitter, base and collector.
- The doping level and width of these three regions play a significant role in the working of the transistor.

**Emitter**

This region is heavily doped compared to the other regions. Since this region transfers the charge carriers to another region for operation, it is called an emitter.

**Base**

This is the least doped region and the most important region because it transfers the carriers from low resistance to high resistance, which will perform the amplification process.

**Collector**

This region is a moderately doped region and has large width and surface, and that is why this region collects the carrier emitted by the emitter region, and hence it is called as a collector.

**Rack Your Brain**

Why does an electronic circuit require a DC supply to work and what happens if we want to operate it using an AC supply?

**Grey Matter Alert!**

Transistor means the transfer of resistance which transfers the signal from low resistance to high resistance.

**Working of transistor:**

- After learning how NPN and PNP transistors are formed, we will now apply some external voltage to the transistor.
- To understand the working of the transistor, let us take a PNP transistor and apply external DC supply voltage to the two junctions which are formed.

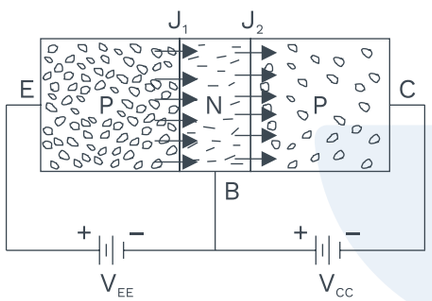



**Fig. 3.3 Movement of Carriers**

- According to the polarities of $V_{EE}$ and $V_{CC}$ shown above, the emitter-base (EB) junction is forward biased, and the collector-base (CB) junction is reverse biased.
- EB-junction is forward biased, which means the width of the junction $J_1$ will reduce, and that of the CB-junction $J_2$ will increase.
- Holes of the P region in the emitter have the same polarity as that of the DC supply and therefore are repelled by the positive terminal of battery $V_{EE}$. As a result, the holes will move from the P to the N region, and since the base region is very lightly doped, a maximum number of holes will pass to the collector.
- Only a few holes recombine with the electrons in the base region resulting in base current $I_B$. For high current gain, the base current should be as low as possible.

**Rack Your Brain**

Why is emitter very heavily doped compared to other regions?

**Grey Matter Alert!**

The ratio of the width of the emitter to the base is 150:1.

**Flow of current in PNP transistor**

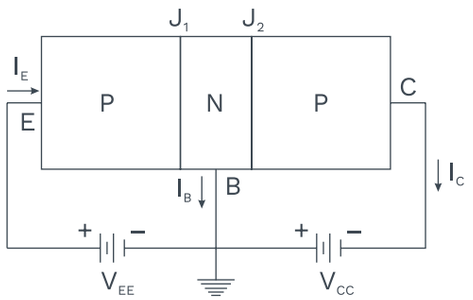



**Fig. 3.4 Current Flow in PNP Transistor**

$I_E$ = Emitter current
$I_B$ = Base current
$I_C$ = Collector current

The relation between these current is given by

$$I_E = I_B + I_C$$

**Working of NPN transistor:**

- The working of the NPN transistor is the same as that of the PNP transistor. The electrons play the same role as being played by holes, but the direction of current is reversed, as shown below.

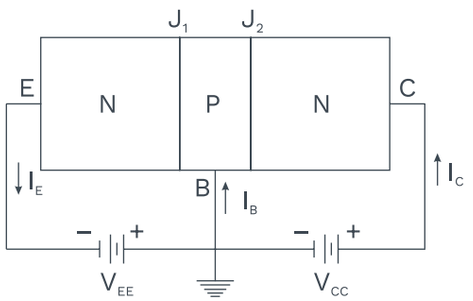



**Fig. 3.5 Current Flowing in NPN Transistor**

**Modes of operation**

Depending upon the polarity of the biasing voltage, we have different modes of operation of transistors, and each mode has different applications.

**Rack Your Brain**

What happens if emitter and collector terminals are interchanged?

**Grey Matter Alert!**

BJT is a device in which we control the flow of electrons between two terminals using a third terminal.

**1)** Both $J_1$ and $J_2$ are forward biased.

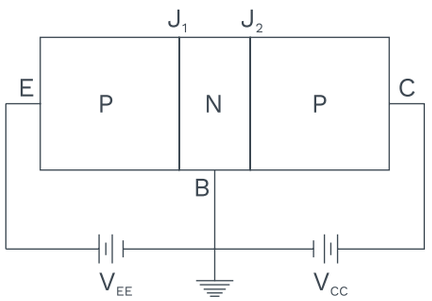



**Fig. 3.6 $J_1$ and $J_2$ Forward Biased**

**$J_1$:** Forward biased
**$J_2$:** Forward biased
**Mode:** Saturation
**Properties:** Very low internal resistance
**Application:** ON switch

**2)** Both $J_1$ and $J_2$ are reverse biased.

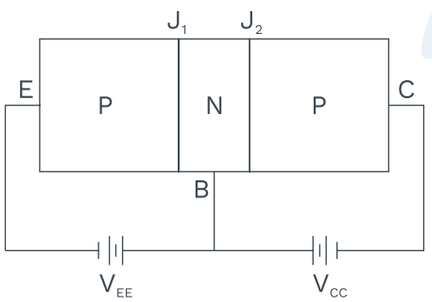



**Fig. 3.7 $J_1$ and $J_2$ Reversed Biased**

**$J_1$:** Reverse biased
**$J_2$:** Reverse biased
**Mode:** Cut-off
**Properties:** Very high internal resistance
**Application:** OFF switch

**Rack Your Brain**

Why is the base region lightly doped and made to be very thin?

**Grey Matter Alert!**

Step-up transformer increases the input voltage applied, but it is not an amplifier.

**3)** $J_1$ is forward biased, and $J_2$ is reverse biased.

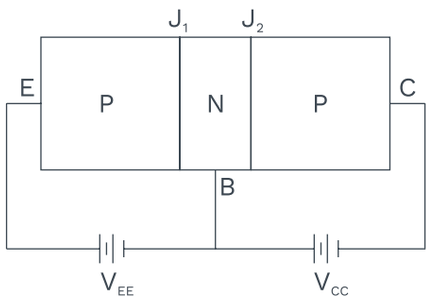



**Fig. 3.8 $J_1$ Is Forward Biased, $J_2$ Is Reverse Biased**

**$J_1$ :** Forward biased
**$J_2$ :** Reverse biased
**Mode:** Active
**Properties:** Excellent transistor action
**Application:** Amplifier

**4)** $J_1$ is reverse biased, and $J_2$ is forward biased.

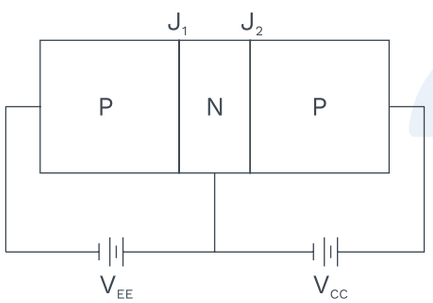



**Fig. 3.9 $J_1$ Is Reverse Biased, $J_2$ Is Forward Biased**

**$J_1$:** Reverse biased
**$J_2$:** Forward biased
**Mode:** Reverse active
**Properties:** Very poor transistor action
**Application:** Attenuator

**Note:** For NPN transistors, the polarity of the voltage source is reversed in every case, whereas properties, mode and applications are the same.

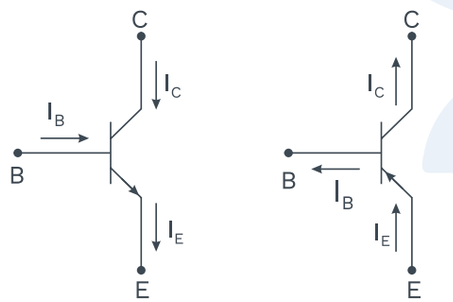
**Rack Your Brain**

Why doping concentrations are different in the three regions of the transistor?

| S No. | $J_1$ | $J_2$ | Mode |
|---|---|---|---|
| 1 | FB | RB | Active |
| 2 | FB | FB | Saturation |
| 3 | RB | RB | Cut-off |
| 4 | RB | FB | Reverse active |

**Symbols of PNP and NPN transistor**

- We have studied the transistor's operation and mode of working but usually, we do not use a three-layer structure to show the transistor. Instead, we use symbols to show transistors.




**NPN transistor** **PNP transistor**

**Fig. 3.10 Symbols of NPN and PNP**

The only difference between the symbols of the PNP and NPN transistor is the arrow.

- In NPN, the arrow is outwards, and in the PNP, the arrow is inwards. It must be noted very carefully that the arrow is always on the emitter terminal.

The direction of the arrow is the direction of the emitter current.

**Grey Matter Alert!**

Every electronic circuit requires a DC power supply for its operation.

## BJT CONFIGURATION

- BJT is a three-terminal device through which the input is applied, and the output is taken the across other two terminals, as shown below.

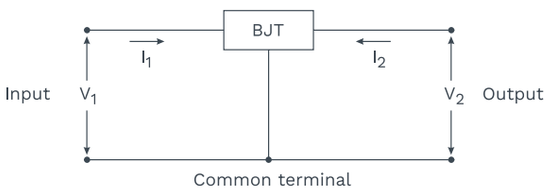



**Fig. 3.11 BJT as a Two Port Device**

- There are three possible configurations of BJT. In every arrangement, one of the terminals out of three (base, emitter, collector) acts as a reference terminal, one as input and the other as output.
  - **i)** Common base configuration (CB-configuration)
  - **ii)** Common emitter configuration (CE-configuration)
  - **iii)** Common collector configuration (CC-configuration)

**Common base configuration:**

- In the common base configuration for the PNP transistor, the base is the common terminal. Emitter-base is the input port, and collector-base is the output port, as shown below.

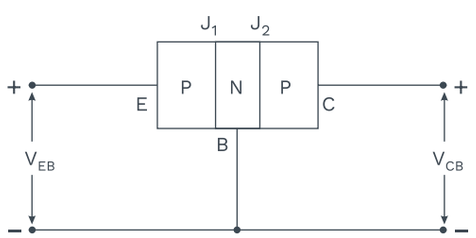



**Fig. 3.12 Common Base Biasing**

- We can see that the input is given across the E-B port, and the output is taken across the C-B port.

**Rack Your Brain**

Why is common base configuration called as constant current source?

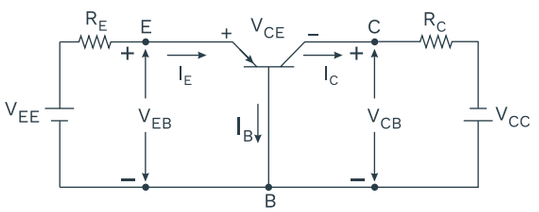



**Fig. 3.13 Circuit for Common Base Configuration**

- The operation of the transistor is the same as we studied in the previous section.

**Input characteristics**

- The input characteristics show how the input current at the input port varies, with the input voltage with the output voltage held constant.

  In the common base configuration, $V_{EB}$ is the input voltage, and $I_E$ is the input current and keeps $V_{CB}$ as the output voltage constant.

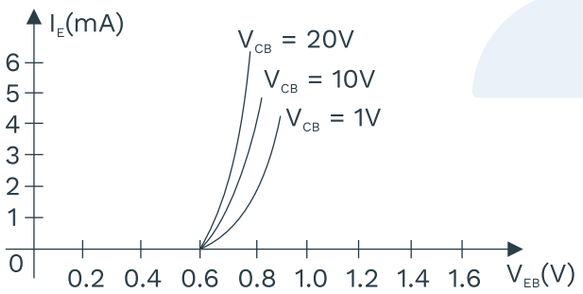



**Fig. 3.14 Input Characteristics of CB**

- It is clearly seen from the above figure that as $V_{CB}$ increases, emitter current also increases.
- When the voltage $V_{CB}$ is 20V, the current is greater than 6 mA, and it decreases as $V_{CB}$ decreases.

**Output characteristics**

The output characteristics of BJT are essential to understand, and it is drawn between the output current $I_C$ and the output voltage $V_{CB}$.

**Rack Your Brain**

Why is transformer not used as an amplifier?

**Grey Matter Alert!**

Two back-to-back PN diodes cannot function as a single BJT because joining two diodes will result in larger base width than emitter and collector regions. For a transistor, the base width must be as small as possible.

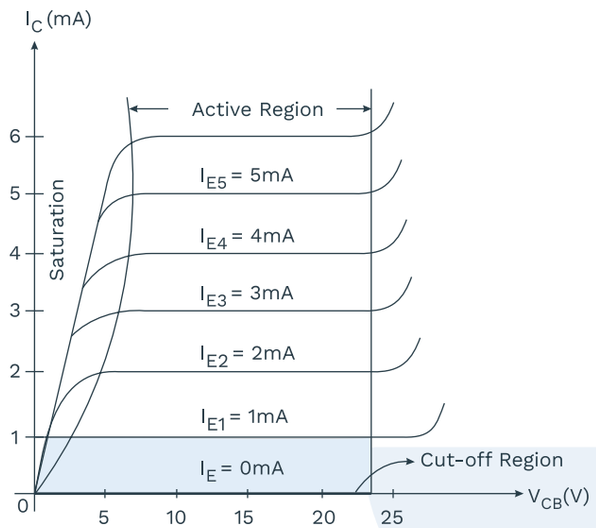



**Fig. 3.15 Output Characteristics of CB**

### Active region

- When junction $J_1$ is forward biased and $J_2$ is reverse biased, the transistor operates in the active region.
- It is the most sensitive region in the characteristics as output current responds more sensitively to an input signal.
- In this region, the transistor is mainly used as an amplifier; the biasing of the transistor should be done in such a way that transistor should operate in an active region only.

### Saturation region

- The region to the left is the ordinate $V_{CB}$ = 0 and above $I_E$ = 0. In this region, both the emitter and the collector junction, i.e. $J_1$ and $J_2$, are forward biased. Whenever the transistor enters into the saturation region, it can be used as an ON switch.

### Cut-off region

- When both the junctions $J_1$ and $J_2$ are reverse biased, the transistor is said to be in the cut-off region. Therefore, the region below $I_E$ =0 is the cut-off region.
- The transistor is mainly used as an OFF switch in the cut-off region.

**Rack Your Brain**

How does a transistor amplify a low voltage signal?

**Grey Matter Alert!**

A transistor is a triode that exists in two forms.

**Current gain**

- Current gain is defined as the ratio of the output current to the input current.

In common base, the output current is $I_C$, and the input current is $I_E$.

$$\text{Current gain} = \frac{\text{Output current}}{\text{Input current}} = \frac{I_C}{I_E} = \alpha$$

Current gain of the common base amplifier is always less than unity, i.e. $\alpha < 1$.

**Example 1:**

A BJT can be used as an amplifier in:

**a)** Saturation region
**b)** Cut-off region
**c)** Active region
**d)** Any region

**Sol:**

A BJT can be used as an amplifier in the active region.

Hence, the correct option is (C).

**Example 2:**

A common base amplifier has an emitter current of 100 mA and a collector current of 98 mA. The current gain is:

**a)** 0.98
**b)** 0.49
**c)** 98
**d)** 100

**Sol:**

Current gain, $\alpha = \frac{I_C}{I_E}$

Input current, $I_C$ = 98 mA

Current, $I_E$ = 100 mA

Current gain $\alpha = \frac{I_C}{I_E} = \frac{98}{100} = 0.98$

Hence, the correct option is (A).

**Previous Year's Questions**

Match the following:

List-I

**I)** The current gain of a BJT will be increased.
**II)** The current gain of a BJT will be reduced.
**III)** The break-down voltage of a BJT will be reduced.

List-II

**a)** The collector doping concentration is increased.
**b)** The base width is reduced.
**c)** The emitter doping concentration to base doping concentration ratio is reduced.
**d)** The base doping concentration is increased, keeping the ratio of the emitter doping concentration to base doping concentration constant.
**e)** The collector doping concentration is reduced.

**Example 3:**

A BJT can be used as an ON switch in which region?

**a)** Cut-off region **b)** Active region

**c)** Saturation region **d)** Reverse active

**Sol:**

A BJT can be used as an ON switch in the saturation region, whereas as an OFF switch in the cut-off region.

Hence, the correct option is (C).

**Common emitter configuration:**

- The common emitter configuration is the most popular configuration of BJT because of its various advantages.
- In CE configuration, the emitter terminal is common to both the input and the output terminals.
- Base-emitter is used as the input port, whereas collector-emitter is used as the output port, as shown in the figure below.

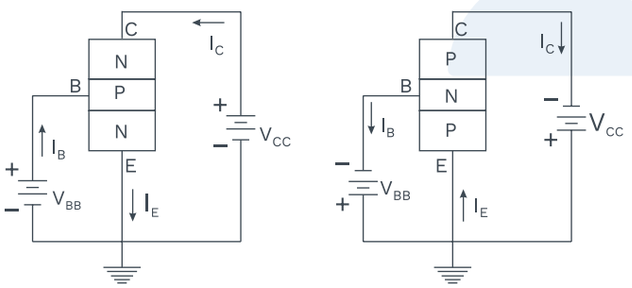



**Fig. 3.16 CE Configuration of NPN and PNP**

Symbolic representations of the above figures are given below.

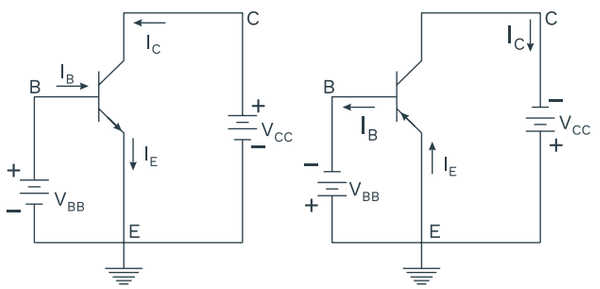



**Fig. 3.17 Biasing of CE Configuration**

**Rack Your Brain**

Why is the current gain of the common base amplifier always less than unity?

- The working of the common emitter configuration is the same as that of the common base.

**Input characteristics**

The input characteristic is drawn between the input voltage $V_{BE}$ and the input current $I_B$, keeping output voltage $V_{CE}$ fixed.

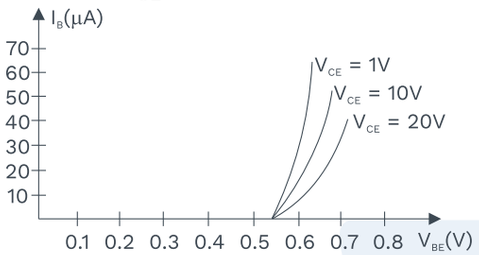



**Fig. 3.18 Input Characteristics of CE Configuration**

- It must be noted carefully that the base current is minimal (in the μA range).
- As $V_{CE}$ increases, the base current decreases.

**Output characteristics**

- It is the most important characteristic of a BJT amplifier.
- The output characteristics are drawn between the output current and the output voltage, keeping the input parameters fixed.
- It shows the variation of the output current ($I_C$) by the output voltage ($V_{CE}$).

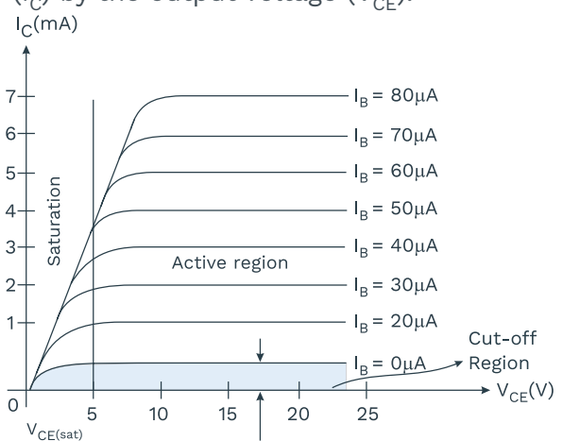



**Fig. 3.19 Output Characteristics of CE Configuration**

**Rack Your Brain**

Why is BJT used as an amplifier in the active region only?

## REGIONS OF OPERATION

**Active region:**

- In the active region, the emitter-base junction is forward biased, whereas the collector-emitter junction is reverse biased.
- The region to the right of $V_{CE(sat)}$ is the active region.
- BJT is used as an amplifier in an active region. This region is preferable for the operation of the transistor.

**Saturation region:**

- In the saturation region, the emitter-base junction is forward biased, and the collector-emitter junction is also forward biased.
- The region to the left of $V_{CE(sat)}$ is the saturation region.
- BJT is used as the ON switch in this region.

**Cut-off region:**

- In the cut-off region, both the junctions are reverse biased.
- The region below $I_B$ = 0μA is the cut-off region. BJT is widely used as an OFF switch in this region.

**Current gain:**

Current gain is defined as the ratio of the output current to the input current. It is denoted by β.

$$\beta = \frac{\text{Output current}}{\text{Input current}} = \frac{I_C}{I_B}$$

$$\beta = \frac{I_C}{I_B}$$

**Relation between α and β:**

$$\beta = \frac{I_C}{I_B}$$

$$\beta = \frac{I_C}{I_E - I_C}$$

$$\frac{I_E - I_C}{I_C} = \frac{1}{\beta}$$

**Grey Matter Alert!**

BJT is a current-controlled device, whereas FET is a voltage-controlled device.

**Previous Year's Questions**

A BJT is said to be operating in the saturation region if:

**a)** Both the junctions are reverse biased.

**b)** Base-emitter junction is reverse biased and base-collector Junction is forward biased.

**c)** Base-emitter junction is forward biased and base-collector junction is reverse-biased.

**d)** Both the junctions are forward biased.

$$\frac{I_E}{I_C} - 1 = \frac{1}{\beta}$$

$$\frac{I_E}{I_C} = \frac{\beta + 1}{\beta}$$

$$\frac{I_C}{I_E} = \frac{\beta}{\beta + 1}$$

but $\alpha = \frac{I_C}{I_E}$

$$\alpha = \frac{\beta}{\beta + 1}$$

Total collector current is given by

$$I_C = \beta I_B + I_{CEO}$$

where $I_{CEO}$ is the collector current with the collector junction reverse biased and the emitter open-circuited.

$$I_C = \beta I_B + (1 + \beta) I_{CBO}$$

where $I_{CBO}$ is the collector current with the collector junction reverse biased and the base open-circuited.

$$I_{CEO} = (1 + \beta) I_{CBO}$$

**Note:** The collector to emitter leakage current should be as low as possible.

The leakage current, which was $I_{CBO}$ in CB configuration, is increased by (1+β) times in CE configuration, which is the biggest drawback of CE configuration.

**Example 4:**

A common emitter (CE) BJT has $\alpha$ = 0.99; what is the value of current gain $\beta$?

**a)** 98 **b)** 99

**c)** 100 **d)** 49

**Sol:**

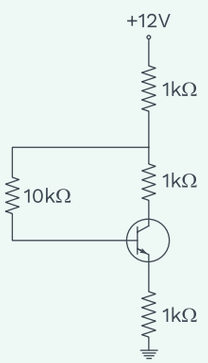
The relation between $\alpha$ and $\beta$ is given by

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{0.99}{1 - 0.99}$$

**Previous Year's Questions**

A transistor having $\alpha$ = 0.99 and $V_{BE}$ = 0.7V is used in the circuit of the figure with a power supply of 12V, the value of the collector current will be:



$$\beta = \frac{0.99}{0.01}$$

$$\beta = 99$$

Hence, the correct option is (B).

**Example 5:**
A common emitter transistor has a current gain of 100, the base current of 100 μA and the collector to base leakage current of 1 μA. Calculate the emitter current.

**Sol:**
Collector current:

$$I_C = \beta I_B + (1+\beta) I_{CBO}$$

$$I_B = 100\mu A$$

$$I_{CBO} = 1\mu A$$

$$I_C = 100 \times 100 \times 10^{-6} + 101 \times 10^{-6}$$

$$I_C = 10 \times 10^{-3} + 0.101 \times 10^{-3}$$

$$I_C = 10.101 \text{ mA}$$

Emitter current

$$I_E = (10.101 + 0.1)\text{mA}$$

$$I_E = 10.201 \text{ mA}$$

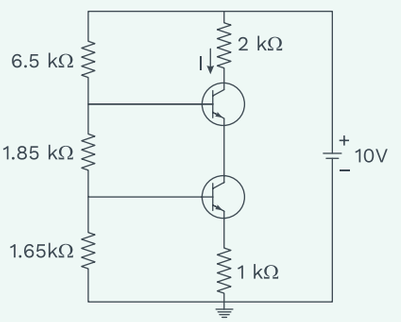
**Example 6:**
Choose the correct order of doping in three regions of the transistor in decreasing order:

**a)** Collector > Emitter > Base
**b)** Emitter > Collector > Base
**c)** Emitter > Base > Collector
**d)** Collector > Base > Emitter

**Sol:**
Since charge carriers flow from emitter to base and then to the collector, the emitter is to be doped heavily so that maximum carriers should reach the collector and have high gain.
The base is the most lightly doped and the collector is also lightly doped.

**Previous Year's Questions**

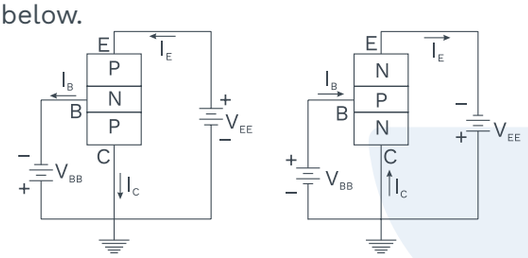
If the transistor in the figure has a high value of β and $V_{BE}$ of 0.65, the current I flowing through 2 kilo-ohms resistance will be _____.



The correct order is Emitter > Collector > Base
Hence, the correct option is (B).

**Common collector configuration:**

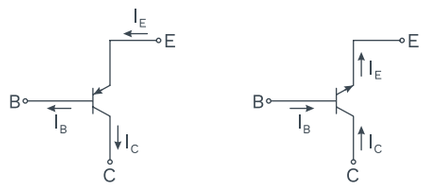
- In a common collector configuration, the collector is the common terminal between input and output.The input is applied across the collector-base port, and output is taken across the collector-emitter port, as shown below.




CC PNP Transistor CC NPN Transistor

**Fig. 3.20 PNP and NPN CC Configuration**

- The working of the PNP and NPN common collector configuration is the same, the only difference being the role played by holes in the PNP transistor is the same as that of the electrons in the NPN transistor.
  The symbolic representation of the above configuration is as shown below:




CC PNP Symbol CC NPN Symbol

**Fig. 3.21 Symbolic Representation of Common Collector Configuration**

**Current gain**

- Current gain is the ratio of the output current and the input current. It is denoted by $\gamma$ in the common collector.

Output current = $I_E$
Input current = $I_B$

**Grey Matter Alert!**

Area of collector, base and emitter is in the order of Collector > Emitter > Base.

$$\gamma = \frac{\text{Output current}}{\text{Input current}} = \frac{I_E}{I_B}$$

$$\gamma = \frac{I_E}{I_B}$$

**Relation between $\alpha$, $\beta$ and $\gamma$**

$$\gamma = \frac{I_E}{I_B} = \frac{I_B + I_C}{I_B}$$

$$\gamma = 1 + \frac{I_C}{I_B}$$

$$\frac{I_C}{I_B} = \beta$$

$$\gamma = 1 + \beta = 1 + \frac{\alpha}{1-\alpha}$$

$$\gamma = 1 + \beta = \frac{1}{1-\alpha}$$

**Example 8:**
Which of the following configuration has maximum current gain?

**a)** Common base **b)** Common emitter
**c)** Common collector **d)** Same for all

**Sol:**
Current gain for common collector is

$$\gamma = \beta + 1$$

So, the common collector has maximum current gain among all configurations.
Hence, the correct option is (C).

**Example 9:**
The current gain '$\alpha$' of BJT transistor is 0.98, find $\gamma$.

**Sol:**

$$\gamma = \frac{1}{1-\alpha} = \frac{1}{1-0.98}$$

$$\gamma = 50$$

**Rack Your Brain**

What is the difference between base transport factor ($\beta^*$) and common emitter current gain ($\beta$)?

## TRANSISTOR CURRENT COMPONENTS

- In the operation of the transistor, there are various current components involved.

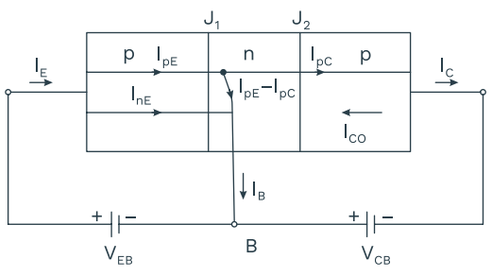



**Fig. 3.22 Current Components in Transistor**

- The emitter current consists of hole current $I_{pE}$ due to holes crossing from the emitter to the base and electron current $I_{nE}$ due to electrons crossing from the base to the emitter.
- The direction of the hole current is the same as that of the conventional current. Some of the holes are combined with the electrons in the base region, and most of the holes reach the collector side.

  $I_{pE} = I_{pC}$ = holes recombining base region

**Emitter injection efficiency:**

It is the ratio of injected carrier current at the emitter junction ($J_1$) to the total current.

$$\gamma = \frac{\text{Injected carrier current at } J_1}{\text{Total emitter current}}$$

$$\gamma = \frac{I_{pE}}{I_{pE} + I_{nE}} = \frac{I_{pE}}{I_E}$$

where, $I_{pE}$ = injected hole diffusion current at the emitter junction

$I_{nE}$ = injected electron diffusion current at emitter junction

**Grey Matter Alert!**

Early effect is the variation of the width of the base in BJT due to the variation in the applied base to collector voltage.

**Transport factor ($\beta^*$):**

$$\beta^* = \frac{\text{Injected carrier current reaching at } J_2}{\text{Injected carrier current at } J_1}$$

$$\beta^* = \frac{I_{pC}}{I_{pE}}$$

In the common base configuration, the large-signal current gain is:

$$\alpha = -\frac{(I_C - I_{CO})}{I_E}$$

**Relation between $\alpha$, $\beta^*$ and $\gamma$:**

$$\alpha = \frac{I_{pC}}{I_E} = \frac{I_{pC}}{I_{pE}} \cdot \frac{I_{pE}}{I_E}$$

$$\alpha = \beta^* \gamma$$

## EARLY EFFECT OR BASE WIDTH MODULATION

- The preferable region for the operation of the transistor is the active region which means that the junction $J_1$ is forward biased and $J_2$ is reverse biased, as shown below.

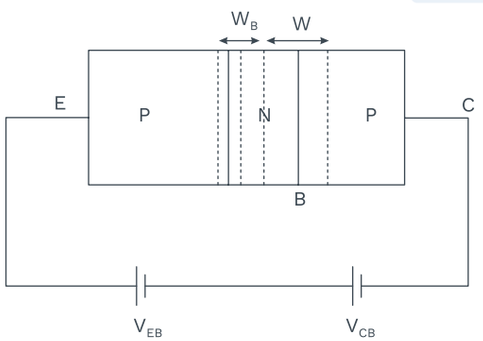



**Fig. 3.23 Early Effect in BJT**

- In the reverse-biased case, the depletion width is very large as compared to the forward bias depletion width.

As we increase the reverse-biased voltage $V_{CB}$, the depletion width increases more and may penetrate to the emitter, which is known as the early effect.

### Previous Year's Questions

Which of the following statements are correct for basic transistor amplifier configurations?

**a)** CB amplifiers have low input impedance and low current gain.

**b)** CC amplifiers have low output impedance and a high current gain.

**c)** CE amplifier has very poor voltage gain but very high input impedance.

**d)** The current gain of CB amplifier is higher than the current gain of CC amplifier.

### Rack Your Brain

What is base width modulation?

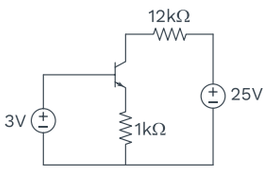
Since base width is changed by reverse bias voltage $V_{CB}$, it is also known as the base width modulation.

**Example 10:**
Early effect generally occurs when the transistor is operating in the:

**a)** Active region
**b)** Cut-off region
**c)** Saturation region
**d)** Reverse active region

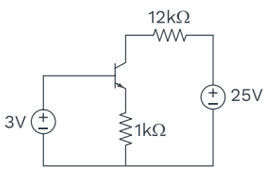
**Sol:**
In the case of early effect, the input junction $J_1$ is forward biased, and the output junction $J_2$ is reverse biased, which is the active region.
Hence, the correct option is (A).

**Example 11:**
An NPN BJT in a CE configuration circuit is shown below with a current gain of infinity and $V_{BE}$ = 0.6V. The BJT operates in which region?




**a)** Saturation
**b)** Cut-off
**c)** Active
**d)** Reverse active

**Sol:**




Applying KVL in the input loop,

$$3 - 0.6 - I_E \times 1k = 0$$

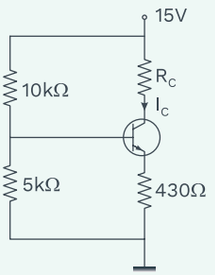
$$I_E = 2.4 \text{ mA}$$

**Grey Matter Alert!**

NPN transistor is faster due to the higher mobility of electrons.

**Previous Year's Questions**

In the circuit of the figure, assume that the transistor is in the active region. It has a large β, and its base-emitter voltage is 0.7V. The value of $I_C$ is




**a)** Indeterminate since $R_C$ is not given
**b)** 1 mA
**c)** 5 mA
**d)** 10 mA

Applying KVL in the outer loop,

$25 - 2.4 \times 10^{-3} \times 12k - V_{CE} - 2.4 \times 10^{-3} \times 1k = 0$

$V_{CE} = -6.2V$ which is not possible

$\therefore$ CE junction is forward biased.

EB junction is also forward biased.

So, the transistor operates in the saturation region.

Hence, the correct option is (A).

**Rack Your Brain**

What are the advantages of biasing in BJT?

| Characteristics | CB Configuration | CE Configuration | CC Configuration |
|---|---|---|---|
| Input impedance | Very low (50 Ω) | Very low (50 kΩ) | Very high (800 kΩ) |
| Output impedance | Very high (1 MΩ) | High (10 kΩ) | Low (50 Ω) |
| Current gain | Less than 1 | High (100) | High (100) |
| Voltage gain | Low (150) | High (500) | Less than 1 |
| Leakage current | Very small | Very large | Very large |
| Application | For high-frequency application | For audio frequency application | For impedance matching |

**BJT Biasing**

- In order to operate BJT in a particular region, we supply DC power through various connections of resistors which are called BJT biasing.
- The AC signal, which is to be amplified using BJT, is applied through some reactive element like a capacitor.
- There are various ways by which we can bias BJT for operating in a particular region.
- If BJT is to be used as an amplifier, it must operate in the active region at some stable

point known as the operating point, also called the Q-point.

**Operating point:**

- In order to use BJT as an amplifier, the transistor should be operating in the active region using some DC power supply (external source).
- For a given value of resistors and power supply, the operating point, which is the most stable point on the characteristics, is obtained and then a time-varying small-signal which is to be amplified is applied.

Consider a common emitter fixed biased circuit (to be discussed later).

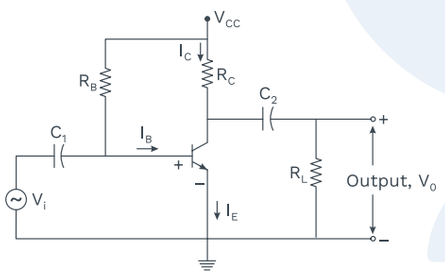



**Fig. 3.24 BJT With AC+DC Supply**

$V_{CC}$ is the power supply, $V_i$ is the small-signal which is to be amplified, $C_1$ is the coupling capacitor which will superimpose the input AC signal over DC source and the output voltage is taken across $R_L$.

We discussed the output characteristics in the previous sections, and we imposed the DC load lines on the output characteristics.

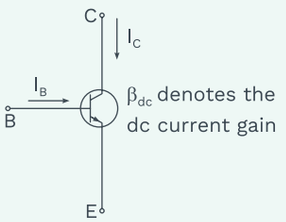
**DC load line:**

For the operating point, we have to do the DC analysis.

For DC analysis, replace the input and output capacitor with an open circuit and input AC signal with a short circuit, as shown below.

**Previous Year's Questions**

If the transistor in the figure is in saturation, then




**a)** $I_C$ is always equal to $\beta_{dc}\ I_B$
**b)** $I_C$ is always equal to $-\beta_{dc}\ I_B$
**c)** $I_C$ is greater than or equal to $\beta_{dc} I_B$
**d)** $I_C$ is less than or equal to $\beta_{dc} I_B$

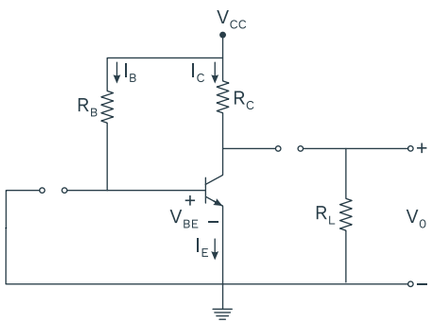



**Fig. 3.25 DC Analysis**

Applying KVL in the input loop, i.e. collector-base loop,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

Applying KVL in the outer loop, i.e. collector-emitter loop,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$I_C = \frac{-V_{CE}}{R_C} + \frac{V_{CC}}{R_C}$$

The equation of $I_C$ represents a linear line with a decreasing slope, i.e. $\frac{-1}{R_C}$ and intercept on the x and y-axis.

x-axis intercept: $I_C = 0$ gives $V_{CE} = V_{CC}$

y-axis intercept: $V_{CE} = 0$ gives $I_C = \frac{V_{CC}}{R_C}$

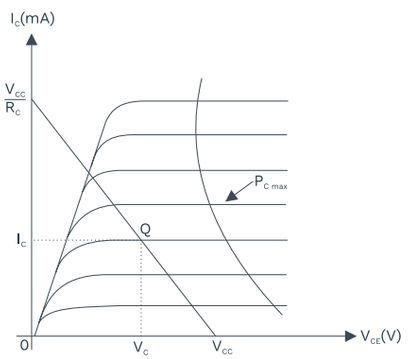


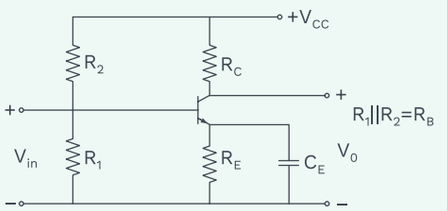

**Fig. 3.26 DC Load Line**

**Grey Matter Alert!**

The DC operating point is the DC steady-state point of the transistor when the AC input signal is zero.

**Previous Year's Questions**

For good stabilised biasing of the transistor of the CE amplifier of the figure, we should have




**a)** $\frac{R_E}{R_B} << 1$ **b)** $\frac{R_E}{R_B} >> 1$

**c)** $\frac{R_E}{R_B} << h_{FE}$ **d)** $\frac{R_E}{R_B} >> h_{FE}$

As clearly seen, the DC load line is the locus of all possible operating points at which BJT is in the active region.

- DC supply $V_{CC}$ and resistances $R_B$, $R_C$ should be chosen such that operating point should lie within the characteristics only.

**Region of operation of transistor**

Calculate the base current ($I_B$) and the collector saturation current ($I_{C(sat)}$).

If $I_B < \frac{I_{C(sat)}}{\beta}$, then transistor is operating in active region.

If $I_B > \frac{I_{C(sat)}}{\beta}$, then transistor is operating in the saturation region.

**Power dissipation**

Power dissipation of transistor is:

$$P_D = I_C V_{CE}$$

$P_D$ is maximum in the active region and minimum in the cut-off and saturation region.

## BIAS STABILISATION

- Semiconductor devices are highly sensitive to temperature variations, which may affect the performance of the transistor.

Collector current of BJT is given by,

$$I_C = \beta I_B + (1 + \beta) I_{CO}$$

$I_{CO}$ is the reverse leakage current which is highly temperature-sensitive and undesirable.

For proper operation of the transistor, the Q-point should not change by the other transistor parameters and temperatures.

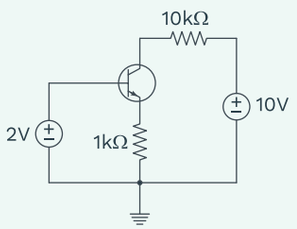
$I_C$ is changed due to the change in $I_{CO}$, $\beta$ and $V_{BE}$.

Using partial differential equation:

$$\Delta I_C = \frac{\partial I_C}{\partial I_{CO}} \times \Delta I_{CO} + \frac{\partial I_C}{\partial V_{BE}} \times \Delta V_{BE} + \frac{\partial I_C}{\partial \beta} \times \Delta \beta$$

**Previous Year's Questions**

For the BJT circuit shown, assume that the $\beta$ of the transistor is huge and $V_{BE}$ = 0.7 V. The mode of operation of the BJT is




**a)** Cut-off
**b)** Saturation
**c)** Normal active
**d)** Reverse active

**Variation due to $I_{CO}$:**

- When $I_{CO}$ changes, the collector current $I_C$ also changes significantly; $I_{CO}$ changes due to the change in the temperature causing $I_C$ to change. It is given by $\frac{\partial I_C}{\partial I_{CO}}$.
- $I_{CO}$ doubles for every 10°C rise in the temperature.

**Variation due to change in $V_{BE}$:**

- $V_{BE}$ is sensitive to the temperature change in $V_{BE}$ that can change the region of operation.

When temperature changes, $V_{BE}$ changes which finally changes the collector current $I_C$. For every 1°C rise, $V_{BE}$ decreases by 2.5 mV

$$\frac{\partial V_{BE}}{\partial T} = -2.5\,\text{mV} / °\text{C}$$

**Variation due to change in $\beta$:**

- Every transistor has a unique current gain $\beta$ value which is highly sensitive to changes in temperature.

Since, $I_C = \beta I_B$

If $\beta$ changes, $I_C$ also changes and is given by $\frac{\partial I_C}{\partial \beta}$.

**Stability factor:**

- Stability factor is the change in the collector current $I_C$ due to the leakage current $I_{CO}$.

$$S = \frac{\partial I_C}{\partial I_{CO}}$$

Collector current, $I_C = \beta I_B + (1 + \beta) I_{CO}$

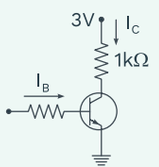
Differentiate partially with respect to $I_C$, assuming $\beta$ and $V_{BE}$ as constant.

$$1 = \frac{\beta \partial I_B}{\partial I_C} + (1+\beta)\frac{\partial I_{CO}}{\partial I_C}$$

$$S = \frac{\partial I_C}{\partial I_{CO}} = \frac{1+\beta}{1-\beta\left(\frac{\partial I_B}{\partial I_C}\right)}$$

**Previous Year's Questions**

Assuming $V_{CEsat} = 0.2V$ and $\beta = 50$, the minimum base current ($I_B$) required to drive the transistor in the figure to saturation is:




**a)** 56 μA **b)** 140 μA
**c)** 60 μA **d)** 3 μA

**Note:** If the numerical value of the stability factor is low, the transistor would be more stable and vice versa.

## BIASING CONFIGURATION

- In order to use BJT in a particular region, proper biasing circuit is required.
- A circuit with the least stability factor will be preferred more.
- For amplification purposes, it should operate in the active region.

**Types of biasing circuits are:**

- Fixed bias circuit
- Collector to base bias circuit
- Self-bias circuit

**Fixed bias circuit**

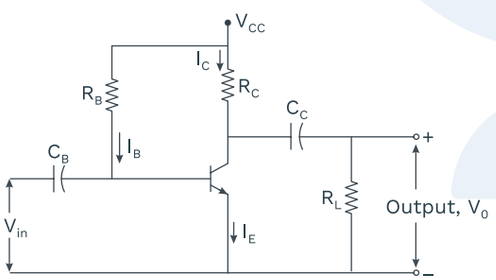



**Fig. 3.27 Fixed Bias Configuration**

In the above configuration, $V_{CC}$ is the voltage supply power to the transistor.
For operating point, DC analysis is required.

**DC analysis**

Replace capacitor by open circuit.

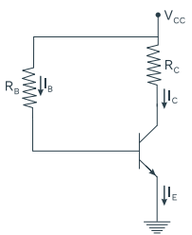
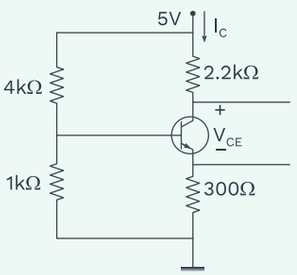



**Fig. 3.28 DC Analysis of Fixed Bias**

### Previous Year's Questions

Assume that the β of the transistor is extremely large and $V_{BE}$ = 0.7 V, $I_C$ and $V_{CE}$ in the circuit shown in the figure are:




**a)** $I_C$ = 1 mA, $V_{CE}$ = 4.7 V
**b)** $I_C$ = 0.5 mA, $V_{CE}$ = 3.75 V
**c)** $I_C$ = 1 mA, $V_{CE}$ = 2.5 V
**d)** $I_C$ = 0.5 mA, $V_{CE}$ = 3.9 V

Applying KVL in input loop,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Applying KVL in output loop,

$$V_{CE} = V_{CC} - I_C R_C$$

$$I_C = \beta I_B = \beta\left(\frac{V_{CC} - V_{BE}}{R_B}\right)$$

$$V_{CE} = V_{CC} - I_C R_C$$

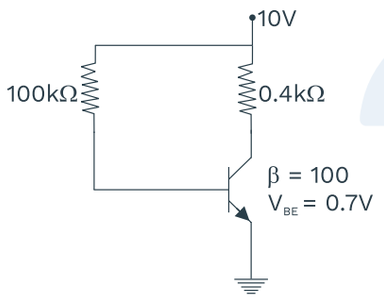
**Stability factor**

- Stability factor for fixed bias circuit is

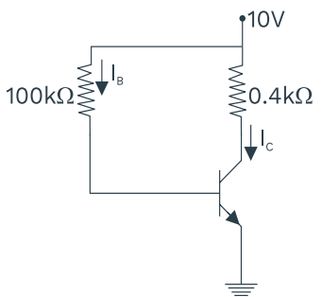
  $S = 1 + \beta$

**Example 12:**

A fixed biased circuit is shown below; Calculate the operating point and stability factor.




**Sol:**




Applying KVL in the input loop,

$10 - 100 \times 10^3 I_B - 0.7 = 0$

**Rack Your Brain**

What is the main disadvantage of fixed bias circuit?

**Grey Matter Alert!**

'A coupling capacitor passes the AC signal from one circuit to another but blocks the DC signal'.

$$I_B = \frac{9.3}{100 \times 10^3}$$

$I_B = 93\mu A$

$I_C = \beta I_B = 100 \times 93 \times 10^{-6}$

$I_C = 9.3mA$

$V_{CE} = 10 - I_C R_C$

$V_{CE} = 10 - 0.4k \times 9.3 \times 10^{-3}$

$V_{CE} = 6.28V$

$I_C = 9.3mA$

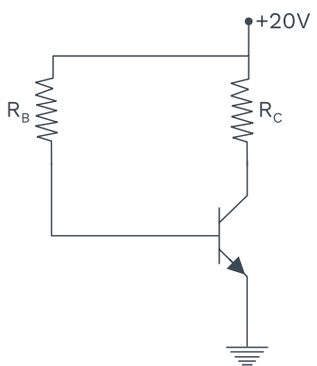
Stability factor for fixed biased circuit is:

$S = 1 + \beta$

$S = 101$

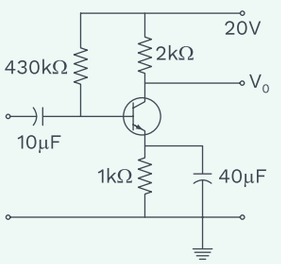
**Note:** It can be clearly seen that the value of the stability factor for fixed bias configuration is very high. So, we need to go for other biasing configurations such as collector to base bias circuits as discussed in the next section.

**Example 13:**
An NPN transistor operating in a common emitter configuration has a current gain of 100. The operating point of transistor is (20mA, 10V). Find $(R_C + R_B)$ in kΩ.




**Previous Year's Questions**

The circuit using a BJT with $\beta = 50$ and $V_{BE} = 0.7$ V is shown in the figure. The base current $I_B$ and collector voltage $V_C$ are, respectively:




**a)** 43 μA and 11.4 V
**b)** 40 μA and 16 V
**c)** 45 μA and 11 V
**d)** 50 μA and 10 V

**Sol:**

Given that,

$I_C = 20mA$

$V_{CE} = 10V$

The transistor is operating in the active region.

$\beta = 100, V_{CC} = 20V$

$V_{BE} = 0.7V, I_C = 20mA$

Applying KVL, $R_C = \frac{V_{CC} - V_{CE}}{I_C}$

$$R_C = \frac{20 - 10}{20 \times 10^{-3}}$$

$R_C = 0.5k\Omega$

and $R_B = \frac{V_{CC} - V_{BE}}{I_B}$

$$I_B = \frac{I_C}{\beta} = \frac{20 \times 10^{-3}}{100}$$

$$R_B = \left(\frac{20 - 0.7}{20 \times 10^{-3}}\right) \times 100$$

$R_B = 96.5k\Omega$

$R_B + R_C = (96.5 + 0.5) \times 10^{-3}$

$R_B + R_C = 97k\Omega$

**Collector to base bias circuit:**

- The stability factor for fixed bias is very high, which means that the fixed bias circuit is very sensitive to the temperature, which is undesirable.

The circuit for the collector to base bias circuit is shown below:

**Rack Your Brain**

What happens if we do not use a coupling capacitor for amplifying a low AC signal?

**Previous Year's Questions**

Introducing a resistor in the emitter of a common emitter amplifier stabilises the DC operating point against variations in

**a)** Only the temperature
**b)** Only the β of the transistor
**c)** Both temperature and β
**d)** None of the above

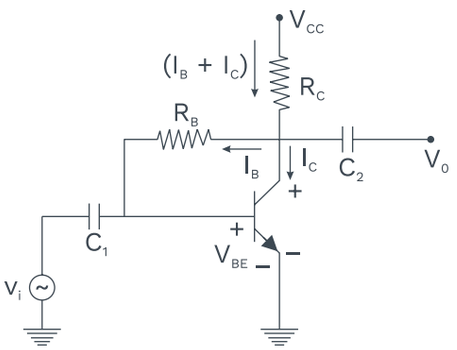



**Fig. 3.29 Collector to Base Bias Configuration**

- In the above circuit, both AC and DC supplies are given.

For DC analysis, remove the capacitor by an open circuit and for AC analysis by a short circuit.

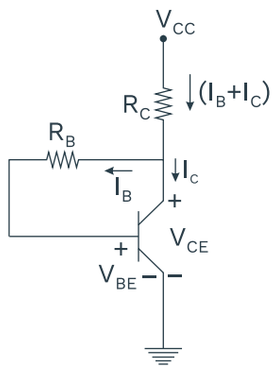



**Fig. 3.30 DC Analysis of Collector to Base Bias**

**Rack Your Brain**

What is the advantage of using an emitter follower circuit?

Applying KVL in the input loop,

$$V_{CC} - (I_B + I_C)R_C - I_BR_B - V_{BE} = 0$$

$$I_C = \beta I_B \text{ (Active region)}$$

$$V_{CC} - I_B(\beta + 1)R_C - I_BR_B - V_{BE} = 0$$

$$I_B[(\beta + 1)R_C + R_B] = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{(\beta + 1)R_C + R_B}$$

Applying KVL in the output loop,

$$V_{CC} - (I_B + I_C)R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - (I_B + I_C)R_C$$

**Stability factor**

- Applying KVL in the input loop,

$$-V_{CC} + (I_B + I_C)R_C + I_BR_B + V_{BE} = 0$$

Differentiate the above equation partially with respect to $I_C$, we get:

$$\left(\frac{\partial I_B}{\partial I_C} + 1\right)R_C + \frac{\partial I_B}{\partial I_C} \cdot R_B = 0$$

$$\frac{\partial I_B}{\partial I_C} = \frac{-R_C}{R_C + R_B}$$

Stability factor, $S = \dfrac{1+\beta}{1-\beta\left(\dfrac{\partial I_B}{\partial I_C}\right)}$

Put $\dfrac{\partial I_B}{\partial I_C} = \dfrac{-R_c}{R_C + R_B}$

$$S = \frac{1+\beta}{1+\beta\left(\dfrac{R_c}{R_C + R_B}\right)}$$

- Clearly, stability for this circuit is less than that of the fixed circuit, which implies that this is more stable with respect to the previous.

**Note:**

- The value of $R_C$ should be large for greater stability of the circuit.
- The base resistance ($R_B$) causes negative feedback, which reduces the gain of the amplifier, which is also undesirable.

**Collector to base bias with emitter follower:**

- In the emitter follower circuit, an emitter resistance $R_E$ is connected, as shown below.

**Rack Your Brain**

Collector current should be insensitive to the variation in $I_{CO}$, $\beta$ and $V_{BE}$.

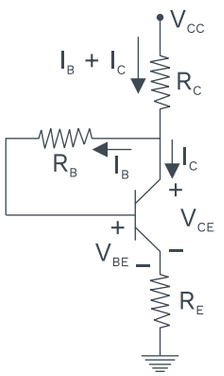



**Fig. 3.31 Collector to Base Bias With Emitter Follower**

Base current is given by,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - (R_C + R_E)I_C$$

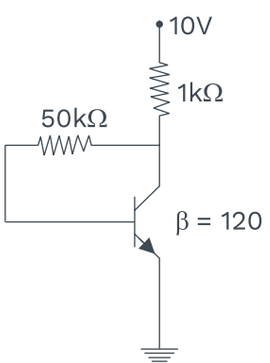
**Advantage of emitter follower circuit**

- Emitter follower circuit is more stable as compared to the previous circuit in which $R_E$ is absent. Stability factor is given by:

$$S = \frac{\beta + 1}{1 + \beta\left(\frac{(R_C + R_E)}{(R_B + R_C + R_E)}\right)}$$

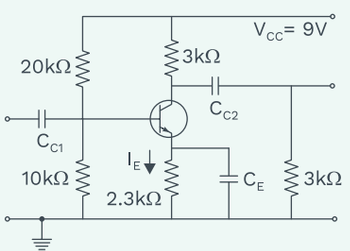
**Example 14:**

A collector to base bias circuit is shown below. Calculate the stability factor.




**Previous Year's Questions**

In the following transistor circuit, $V_{BE}$ = 0.7V, $r_e = 25mV/I_E$, β and all the capacitances are very large.




The value of DC current $I_E$ is:

**a)** 1 mA **b)** 2 mA

**c)** 5 mA **d)** 10 mA

**Sol:**

Stability factor of the collector to base feedback configuration is given by:

$$S = \frac{1+\beta}{1+\beta\dfrac{R_C}{R_B + R_C}}$$

$$S = \frac{1+120}{1+120\times\dfrac{1}{51\times 10^3}}$$

$S = 36$

**Example 15:**

In the above example, if the fixed bias circuit would have been used with the same value of collector and base resistance, calculate the stability factor and compare it with the collector to base bias feedback circuit.

**Sol:**

Stability factor of fixed bias is $1 + \beta$

$= 1 + 120 = 121$

It can be easily seen that fixed bias has a much higher stability factor than collector to base bias circuit.

**Voltage-divider bias:**

- The main disadvantage of the collector to base bias circuit is that we require a large value of $R_C$ to have a more stable operation. For this, we require some other configuration which is voltage-divider bias.

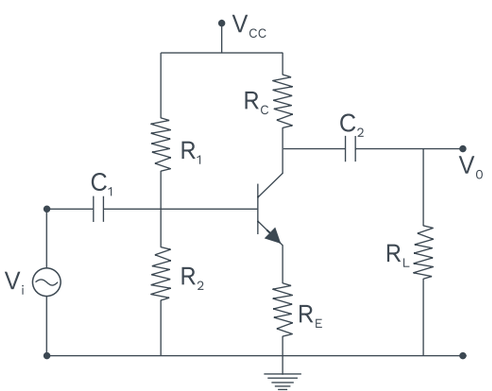



**Fig. 3.32 Voltage Divider Bias Configuration**

- $V_{CC}$ is for power supply
  $V_i$ is input applied

$V_0$ is output taken across load $R_L$
$C_1$, $C_2$ is the coupling capacitor
$R_E$ is the emitter follower resistance

- In order to understand the performance of voltage-divider bias circuit, DC analysis is required.

**DC analysis**

- Replace capacitor with an open circuit and AC voltage source with a short circuit.

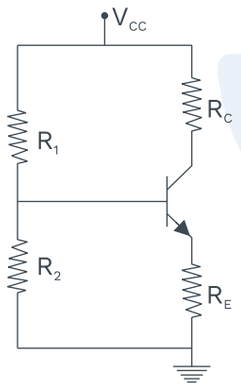



**Fig. 3.33 DC Analysis of Voltage Divider**

- The analysis of the above circuit is done by obtaining Thevenin's equivalent circuit.

**Thevenin's voltage $V_{Th}$**

$R_1$ and $R_2$ forms the voltage divider networks.

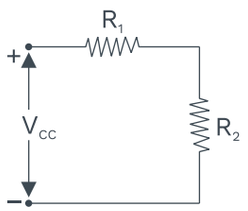



**Fig. 3.34 Thevenin's Circuit**

Voltage across $R_2$ is the Thevenin's voltage which is given by:

$$V_{Th} = \frac{V_{CC}R_2}{R_1 + R_2}$$

**Rack Your Brain**

Why thermal runaway occurs in BJT only and not in FET devices?

**Thevenin's resistance**

$R_1$ and $R_2$ are in parallel

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

The simplified circuit is shown below,

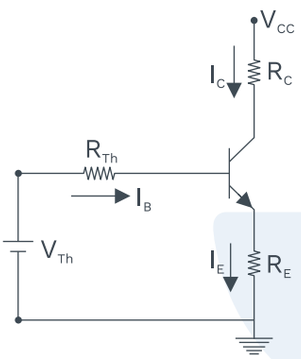



**Fig. 3.35 Simplified Circuit of Voltage Divider**

**Rack Your Brain**

Why is thermal stability important in BJT transistors?

Applying KVL in the input loop, we get:

$$V_{Th} - I_B R_{Th} - V_{BE} - (I_B + I_C)R_E = 0$$

$$I_C = \beta I_B \text{ (Active region)}$$

$$V_{Th} - V_{BE} = I_B[R_{Th} + (\beta + 1)R_E]$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$$

Applying KVL in the outer loop, we get:

$$V_{CC} - I_C R_C - V_{CE} - (I_B + I_C)R_E = 0$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) - I_B R_B$$

**Stability factor**

On partially differentiating the equation of input KVL with respect to $I_C$, and we get:

$$-\frac{\partial I_B}{\partial I_C} R_{Th} - \left(\frac{\partial I_B}{\partial I_C} + 1\right) R_E = 0$$

$$\frac{\partial I_B}{\partial I_C} = \frac{-R_E}{R_B + R_E} \quad \{R_{Th} = R_B\}$$

Stability factor, $S = \dfrac{1+\beta}{1+\beta\dfrac{R_E}{R_B + R_E}}$

- It can be clearly seen from the above equation that the stability factor is independent of the collector resistance '$R_C$', which means that whatever be the value of resistance $R_C$, it will not affect the stability of the given configuration.
- Voltage divider circuit is the most stable circuit among all the configurations.

**Currents in common base configuration:**

- Emitter current in common base configuration is given by:

  $I_E = I_B + I_C + I_{CBO}$

  $I_E$ = Emitter current

  $I_B$ = Base current

  $I_C$ = Collector current

  $I_{CBO}$ = Collector to base leakage current

**$I_{CBO}$:** This current plays a very important role in the performance of BJT. This current is highly sensitive to the temperature change and should be as low as possible.

$I_{CBO}$ doubles for every 10°C rise in temperature.

The total collector is given by:

$I_C = \alpha I_E + I_{CBO}$

**Note:** For proper working of the transistor, the leakage current should be as low as possible.

## THERMAL RUNAWAY

- Semiconductor devices are highly temperature-sensitive devices; when temperature increases, the collector current $I_C$ also increases, which further increases temperature causing high heat in the transistor and the transistor may damage completely.

  $T\uparrow \longrightarrow I_C\uparrow \longrightarrow T\uparrow\uparrow \longrightarrow$ Damaging of BJT

If $T_j$ = Junction temperature of collector



**Rack Your Brain**

How the resistance of the thermistor varies with the temperature?

$T_A$ = Ambient temperature

$P_D$ = Power dissipation across collector junction

Then, thermal resistance ($\theta$) is:

$$\theta = \frac{T_j - T_A}{P_D} \text{ (°C/watt or °K/watt)}$$

- Thermal resistance is analogous to current limiting resistance.

Thermal runaway is an unwanted phenomenon in BJT amplifiers, and it should be avoided, so the question is how to avoid thermal runaway?
A transistor is said to be thermally stable if:

$$\frac{\partial P_c}{\partial T_j} \leq \frac{\partial P_D}{\partial T_j} = \frac{1}{\theta}$$

$\frac{\partial P_c}{\partial T_j}$ = Rate of release of heat

$\frac{\partial P_D}{\partial T_j}$ = Rate of dissipation of heat

For thermal stability, $V_{CE} < \frac{V_{CC}}{2}$

**Compensation method:**
The collector current is highly temperature-sensitive and changes with temperature. In order to stabilise collector current $I_C$, we have to use some temperature-dependent resistors such as thermistors, sensistors and devices such as a diode. We will discuss temperature compensation using these elements one by one.

**1) Diode compensation**

In this case, the diode is operated in reverse biased mode.

**Rack Your Brain**

How is current mirror circuit so useful in powering multiple matched transistors?

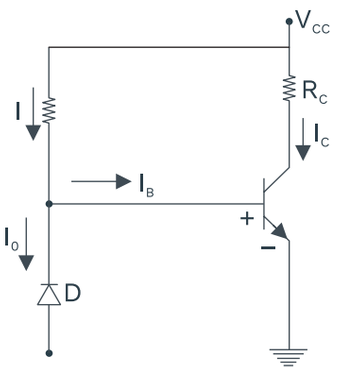



**Fig. 3.36 Diode Compensation Circuit**

- $I_0$ is the reverse saturation current of the diode, and if the diode and transistor are made up of the same material, then $I_0$ and $I_{CO}$ are nearly equal to each other, causing cancellation.

$I_C = \beta I - \beta I_0 + (1 + \beta) I_{CO}$

If $I_0 = I_{CO}$ and if $\beta$ is high $\beta >> 1$

$I_C \approx \beta I \approx \beta I_B$

**2) Thermistor compensation**

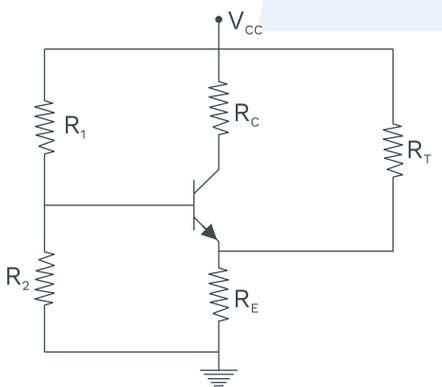



**Fig. 3.37 Thermistor Compensation Circuit**

- $R_T$ is nothing but a thermistor which is a temperature-dependent resistor. As temperature increases, $R_T$ decreases, causing leakage current to flow through $R_T$.

**Example 16:**
Which of the following statement is not correct about thermistors?
**a)** It is used for temperature measurement
**b)** It is highly sensitive
**c)** It is made up of semiconductor
**d)** It is made up of metals

**Sol:**
A thermistor is a temperature sensor and is highly sensitive. It is made up of semiconductors and not metal.
Hence, the correct option is (D).

**Example 17:**
The condition for thermal stability of BJT is:

**a)** $V_{CE} < \frac{V_{CC}}{2}$ **b)** $V_{CE} > \frac{V_{CC}}{2}$

**c)** $V_{CE} = \frac{V_{CC}}{2}$ **d)** $V_{CE} \geq \frac{V_{CC}}{2}$

**Sol:**
The condition to avoid thermal runaway in BJT is:

$$V_{CE} < \frac{V_{CC}}{2}$$

So, condition for thermal stability is:

$$V_{CE} < \frac{V_{CC}}{2}$$

Hence, the correct option is (A).

**Rack Your Brain**

What is the main disadvantage of a normal current mirror and how can it be improved using Widlar current mirror?

## CURRENT MIRROR CIRCUIT

- After learning about BJT biasing, the next question is how to use BJT in any integrated circuit. Nowadays, millions of transistors are fabricated on a single silicon chip.
- Transistors usually require a DC power supply to work. So, if we have to fabricate millions of transistors, we require millions of power supply. But is it practical to give millions of power supplies for a single chip?

The answer is NO. So, the next question is how to power millions of transistors using a single power supply?

- The answer is current mirror circuit. Using a current mirror circuit, we can power a number of transistors using a single power supply.

**What is a current mirror?**

- Current mirror is the circuit in which the same collector current flows through two matched transistors.

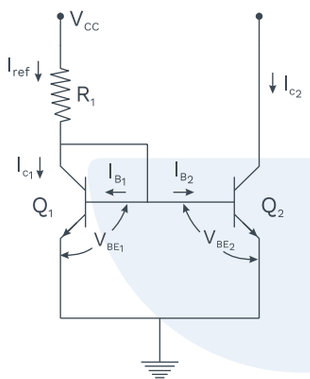



**Fig. 3.38 Current Mirror Circuit**

- The base and emitter of two transistors, $Q_1$ and $Q_2$, are tied together.
- The base current of transistors should be very low, which means that the current gain $\beta$ of the transistor should be very large.

Collector current, $I_C = I_S e^{V_{BE}/V_T}$

Where $I_S$ = saturation current

$$I_{C_1} = I_{S_1} e^{V_{BE1}/V_T}$$

$$I_{C_2} = I_{S_2} e^{V_{BE2}/V_T}$$

Since two transistors are matched, $I_{S1} = I_{S2}$

$$\frac{I_{C_1}}{I_{C_2}} = e^{(V_{BE1}-V_{BE2})/V_T}$$

Applying KCL at the collector node of transistor $Q_1$

$$I_{ref} = I_{C_1} + I_{B_1} + I_{B_2}$$

Transistors are identical

$$V_{BE_1} = V_{BE_2}$$

$$I_{C_1} = I_{C_2} \Rightarrow I_{B_1} = I_{B_2}$$

$$\beta_1 = \beta_2 = \beta$$

$$\therefore \quad I_{ref} = I_{C_1} + 2I_{B_1}$$

Let $I_{C_1} = I_{C_2} = I_C$ and $I_{B_1} = I_{B_2} = I_B$

$$I_{ref} = I_C + 2I_B$$

Since the transistors are operating in the active region,

$$I_B = \frac{I_C}{\beta}$$

$$I_{ref} = I_C + \frac{2}{\beta} I_C$$

$$I_C = \frac{\beta}{(\beta + 2)} I_{ref}$$

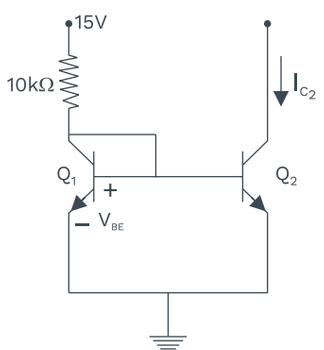
Applying KVL in the input loop of transistor $Q_1$,

$$V_{CC} - I_{ref} R_1 - V_{BE} = 0$$

$$I_{ref} = \frac{V_{CC} - V_{BE}}{R_1}$$

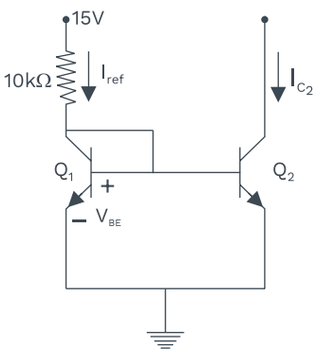
**Example 18:**

A current mirror circuit is shown below, which has a current gain of 200;, find current $I_{C2}$ in transistor $Q_2$ if $V_{BE}$ = 0.7V for the transistor.



**Sol:**

Given that, current gain $\beta = 200$, $V_{BE} = 0.7$ V




Applying KVL at input of $Q_1$,

$$15 - 10k\, I_{ref} - 0.7 = 0$$

$$I_{ref} = 1.43 \text{ mA}$$

$$I_{C_2} = \frac{\beta}{\beta + 2} \times 1.43 \times 10^{-3} = \frac{200}{202} \times 1.43 \times 10^{-3}$$

$$I_{C2} = 1.41 \text{ mA}$$

**Widlar current mirror:**

The main disadvantage of the current mirror is the resistance $R_1$ which should be very high and is not fabricated very easily.

So we have to modify the current mirror circuit as shown below.

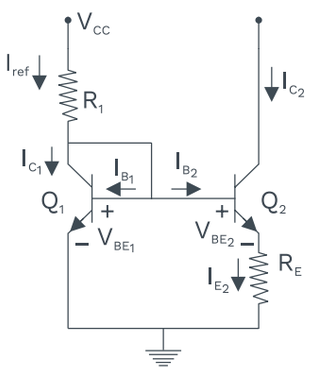



**Fig. 3.39 Widlar Current Mirror Circuit**

Ratio of collector current is,

$$\frac{I_{C_1}}{I_{C_2}} = e^{(V_{BE_1} - V_{BE_2})/V_T}$$

Taking natural log on both sides, we get:

$$\ln\left(\frac{I_{C_1}}{I_{C_2}}\right) = \ln e^{(V_{BE_1} - V_{BE_2})/V_T}$$

$$\frac{V_{BE_1} - V_{BE_2}}{V_T} = \ln\left(\frac{I_{C_1}}{I_{C_2}}\right)$$

$$V_{BE_1} - V_{BE_2} = V_T \ln\left(\frac{I_{C_1}}{I_{C_2}}\right)$$

$$V_{BE_1} = V_{BE_2} + V_T \ln\left(\frac{I_{C_1}}{I_{C_2}}\right)$$

Applying KVL in the base-emitter loop, we get:

$$V_{BE_1} - V_{BE_2} - I_{E_2} R_E = 0$$

$$V_{BE_1} = V_{BE_2} + (I_{B_2} + I_{C_2}) R_E$$

$$I_{C_2} = \beta I_{B_2}$$

$$V_{BE_1} - V_{BE_2} = \left(\frac{1}{\beta} + 1\right) I_{C_2} R_E$$

$$R_E = \frac{\beta}{(\beta + 1)} \times \frac{V_T}{I_{C_2}} \ln\left(\frac{I_{C_1}}{I_{C_2}}\right)$$

$$I_{ref} = I_{C_1} + I_{B1} + I_{B_2}$$

$$I_{ref} = I_{C_1}\left(1 + \frac{1}{\beta}\right) + I_{B_2}$$

$$I_{ref} = I_{C_1}\left(1 + \frac{1}{\beta}\right) + \frac{I_{C_2}}{\beta}$$

Due to resistance $R_E$, $I_{C_2} << I_{C_1}$

$$I_{C_1} = \frac{\beta}{(\beta + 1)} I_{ref}$$

$$I_{ref} = \frac{V_{CC} - V_{BE}}{R_1}$$

If $\beta$ is very large, $I_{C_1} = I_{ref}$

**Multiple transistors current mirror:**

- We already discussed that silicon chip contains millions of transistors that need power for operation.

Consider N transistors connected as shown below.

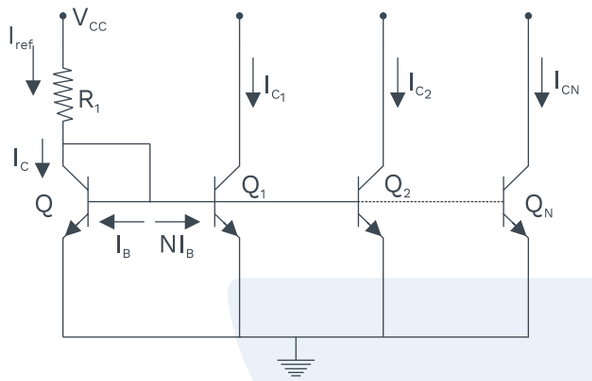



**Fig. 3.40 Multiple Current Mirror Circuits**

Since all the transistors are identical,

$$I_{B_1} = I_{B_2} = I_{B_3} = I_B$$

Applying KCL at collector node of transistor Q,

$$I_{ref} = I_C + \frac{I_C}{\beta}(N+1)$$

$$I_{ref} = I_C\left(1 + \frac{(1+N)}{\beta}\right)$$

$$I_C = I_{ref}\frac{\beta}{\beta + 1 + N}$$

In the initial circuit, two transistors are used, N = 1

$$I_C = I_{ref}\frac{\beta}{\beta + 2}$$

**Note:** In all the current mirror circuits, single power supply is used.

**Wilson current mirror:**

- All the discussed current mirrors are suffering from low output impedance. In this section, we discuss a special current mirror known as the Wilson current mirror, which has a very high output impedance and output current nearly as same as the current through the power supply.

The circuit diagram for the Wilson current mirror is shown below.

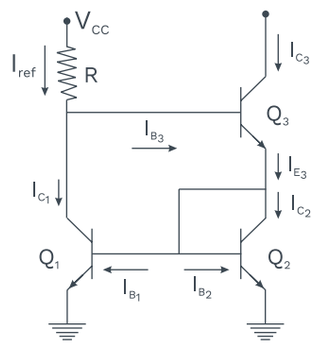



**Fig. 3.41 Wilson Current Mirror Circuit**

$I_{C1} = I_{C2}$ and $I_{B1} = I_{B2} = I_B$

Applying KCL at collector of $Q_2$, we get:

$$I_{E3} = 2I_B + I_{C2}$$

Also, $I_{E3} = I_{B3} + I_{C3}$

$$I_{E3} = \left(\frac{2}{\beta} + 1\right) I_{C2} = I_{C3}\left(1 + \frac{1}{\beta}\right) \qquad \left\{ I_{B1} = I_{B2} = I_B = \frac{I_{C2}}{\beta} \right\} \quad \left\{ I_{B3} = \frac{I_{C3}}{\beta} \right\}$$

or, $$I_{C3}\left(1 + \frac{1}{\beta}\right) = I_{C2}\left(1 + \frac{2}{\beta}\right)$$

$$I_{C3} = \left(\frac{\beta + 2}{\beta + 1}\right) I_{C2}$$

Since, $I_{C1} = I_{C2}$

Thus, $$I_{C3} = \left(\frac{\beta + 2}{\beta + 1}\right) I_{C1}$$

$$I_{ref} = I_{C1} + I_{B3} = \left(\frac{\beta + 1}{\beta + 2}\right) I_{C3} + \frac{I_{C3}}{\beta} \qquad \left\{ I_{C1} = \left(\frac{\beta + 1}{\beta + 2}\right) I_{C3} \right\}$$

$$I_{ref} = I_{C3}\left(\frac{\beta^2 + 2\beta + 2}{\beta^2 + 2\beta}\right)$$

$$I_{C3} = \left(\frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2}\right) \cdot I_{ref}$$

## CHAPTER SUMMARY



- BJT is a three-terminal device having two junctions, hence called a bipolar device.
- Transistors always require a DC power supply to work.
- The amplification signal to the transistor is generally an AC signal.
- Emitter is the highly doped region in BJT, whereas base is lightly doped.
- The width of the collector region is highest, whereas that of the base region is lowest.
- The working of the PNP and NPN transistors is the same, except the direction of current is reversed in both transistors.
- Emitter current is $I_E = I_B + I_C$.
- The current in the PNP transistor is mainly due to holes, whereas that in NPN is due to electrons.
- BJT is mainly used as an open switch, a closed switch and an amplifier.
- BJT is used as an amplifier in the active region only.
- In the symbol of PNP and NPN transistors, the direction of arrows indicates the emitter current.
- The current gain of the common base BJT is $\alpha$ and is the ratio of $I_C/I_E$.
- The current gain of the common emitter type configuration is $\beta$, and its value is $I_C/I_B$.
- Input characteristics are the variation of input current with input voltage.
- Output characteristics are the variation of output current with output voltage.
- Common collector configuration is used in impedance matching.
- Common emitter provides 180° phase shift between input and output.
- Emitter efficiency is the measure of how much charge carriers reach the collector from the emitter.
- Early effect is also known as base width modulation.
- Proper biasing is necessary to operate transistors at the operating point.
- Collector current of the transistor is very sensitive to variation in temperature.
- Three types of biasing circuits are used: fixed bias, collector to base bias and voltage-divider bias.
- Thermal runaway may damage the transistor due to its high temperature during operation.
- Diode, thermistor and sensistor are used for temperature compensation.
- Current mirror is a circuit used for biasing multiple transistors using a single power supply.

## PRACTICE QUESTIONS

**Q1** Which of the following configuration has the lowest input resistance?
a) Common base b) Common emitter
c) Common collector d) None of the above

**Q2** Which of the following configuration provides a negative phase shift between input and output?
a) Common base b) Common emitter
c) Common collector d) None of the above

**Q3** The input impedance of common collector configuration is of the order of
a) 100Ω b) 1kΩ c) 1MΩ d) 50kΩ

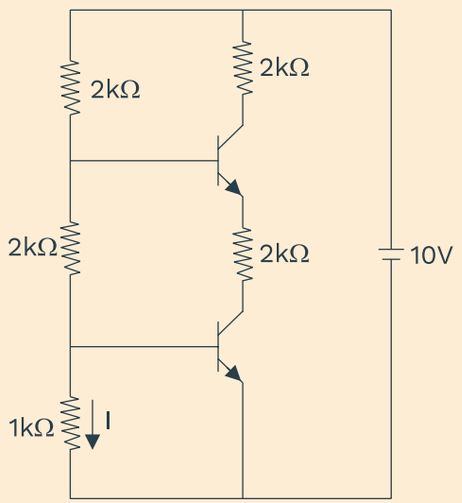
**Q4** The configuration of cascode amplifier is
a) CE – CE b) CE – CB c) CC – CB d) CC – CC

**Q5** The bandwidth of a five-stage amplifier with each stage having bandwidth of 1 KHz is
a) 686 Hz b) 486 Hz c) 586 Hz d) 386 Hz

**Q6** Consider a Darlington pair amplifier each having a current gain of 200. The overall current gain is
a) 40200 b) 40100 c) 400 d) 40000

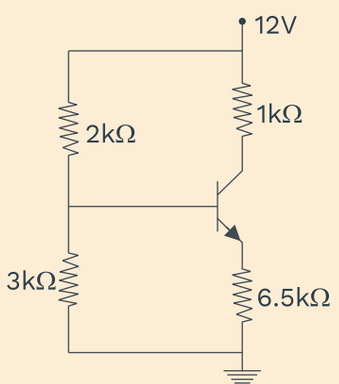
**Q7** A BJT is used as amplifier when. ($J_1$ = input junction, $J_2$ = output junction)
a) $J_1$ is FB, $J_2$ is FB b) $J_1$ is FB, $J_2$ is RB
C) $J_1$ is RB, $J_2$ is RB d) $J_1$ is RB, $J_2$ is FB

**Q8** **A transistor circuit is shown below having a very large current gain and $V_{BE}$ = 0.7V; what is the current through 1kΩ resistor?**




**a) 2 mA** **b) 1 mA** **c) 0.5 mA** **d) 5 mA**

**Q9** **The circuit below has a very large current gain and $V_{BE}$ = 0.7V. The value of $I_C$ is**




**a) 4 mA** **b) 3 mA** **c) 2 mA** **d) 1 mA**

**Q10** Which of the following configuration of BJT is used in audio frequency applications?

a) Common base
b) Common emitter
C) Common collector
d) None of the above

**Q11** If base current is $I_B > \frac{I_{Csat}}{\beta}$, the transistor is said to be operating in

a) Active region
b) Cut-off region
c) Saturation region
d) Reverse active

**Q12** Power dissipation is maximum in which region?

a) Active region
b) Saturation region
c) Cut-off region
d) Reverse active

**Answer Key:**

| | | | | | |
|---|---|---|---|---|---|
| 1) a) | 2) b) | 3) c) | 4) b) | 5) d) | 6) a) |
| 7) b) | 8) a) | 9) d) | 10) b) | 11) c) | 12) a) |